5 Case Assessment and Treatment Planning

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CHAPTER OUTLINE

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The process of case selection and treatment planning begins after a clinician has diagnosed an endodontic problem. The clinician must determine whether the patient's oral health needs are best met by providing endodontic treatment and maintaining the tooth or by advising extraction. The use of cone beam computed technology (CBCT), rotary instruments, ultrasonics, and microscopy, as well as new materials, has made it possible to retain teeth that previously would have been extracted. In addition, even teeth that have failed initial endodontic treatment can often be successfully retreated using nonsurgical or surgical procedures.

Increased knowledge of factors involved in intraoperative and postoperative pain have led to new pain preventive strategies. For example, recognition of the importance of anxiety control, premedication with a nonsteroidal antiinflammatory drug (NSAID) or acetaminophen, profound local anesthesia, and appropriate occlusal adjustment enables clinicians to complete endodontic procedures without intraoperative or posttreatment pain.

The scope of endodontics has changed; the clinician now has more viable options than ever before. In recent years, there is increasing evidence to support the implementation of regenerative procedures for some immature teeth, vital pulp therapy, and replantation of teeth with failed nonsurgical or surgical endodontic procedures. Treatment planning must now include those options as well as nonsurgical or surgical endodontics.

Questions concerning tooth retention and possible referral can only be answered after a complete patient evaluation. The evaluation must include assessment of medical, psychosocial, and dental factors as well as consideration of the relative complexity of the endodontic procedure. Although most medical conditions do not contraindicate endodontic treatment, some can influence the course of treatment and require specific modifications. Comprehensive texts are available that review the subject of dental care for the medically compromised patient.[15,51,118] The American Academy of Oral Medicine (Edmonds, WA) has an excellent website (www.aaom.com) that can be used to elicit information about medically compromised patients.

The American Dental Society of Anesthesiology (ADSA) has produced a valuable source of information, "Ten Minutes Saves a Life" application, which is available for most common smartphone platforms. Its goal is to optimize patient safety and outcomes during in-office medical emergencies (see https://www.adsa-arf.org/tenminutes).

Perhaps the most important advice for a clinician who plans to treat a medically compromised patient is to be prepared to communicate with the patient's physician. The proposed treatment can be reviewed, and medical recommendations should be documented. Fig. 5.1 depicts a sample medical consultation letter that can be modified as necessary.

The American Society of Anesthesiologists (ASA; Park Ridge, IL) Physical Status Classification system is commonly used to express medical risk (Box 5.1). The ASA classification system remains the most widely used assessment method for preanesthetic patients, despite some inherent limitations to its use as a peritreatment risk predictor. This classification system is a generally accepted useful guide for pretreatment assessment of relative risk but does not advise appropriate treatment modifications. The clinician should go beyond the classification system and gather more information from the patient and physician, including the

Michael White, MD
1 Walker Street
Brown City, OK

Dear Dr. White,

Your patient, Ms. Mary Smith, presented for consultation on August 10, 2009, concerning tooth #19. The tooth is asymptomatic at this time, but a small (4 mm x 3 mm), well-circumscribed periradicular radiolucency associated with the palatal root was noted on radiographic examination. The tooth was tested for vitality using thermal and electrical modalities and found to be nonvital, indicating an odontogenic cause for the lesion. The tooth will require endodontic treatment in order to be maintained. The prognosis for nonsurgical endodontic treatment in this case is good. I anticipate that her dental medication plan would include lidocaine with epinephrine for anesthesia and ibuprofen for postoperative pain control.

In a review of the patient's medical history she noted that she is being treated for a malignancy of the thyroid gland and is undergoing radiation therapy. She was unable to provide more specific information about her treatment.

I would appreciate information regarding her ability to undergo endodontic treatment at this time. Please call me if there is further information that you would require concerning the dental treatment to be provided. Thank you.

Yours truly,

Peter Jones, DDS

Fig. 5.1 Sample medical consultation letter.

Box 5.1 American Society of Anesthesiologists Physical Status Classification System

ASA I: A normal healthy patient healthy, non-smoking, no or minimal alcohol use.
ASA II: A patient with mild systemic disease, mild diseases only without substantive functional limitations. Examples include (but not limited to): current smoker, social alcohol drinker, pregnancy, obesity (30 < BMI < 40), well-controlled DM/HTN, mild lung disease.
ASA III: A patient with severe systemic disease, substantive functional limitations; one or more moderate to severe diseases. Examples include (but not limited to): poorly controlled DM or HTN, COPD, morbid obesity (BMI ≥40), etc.
ASA IV: A patient with severe systemic disease that is a constant threat to life. Examples include (but not limited to): recent (<3 months) MI, CVA, TIA, or CAD/stents, etc.
ASA V: A moribund patient who is not expected to survive without the operation.
ASA VI: A declared brain-dead patient whose organs are being removed for donor purposes.

ASA, American Society of Anesthesiologists; *BMI,* body mass index; *CAD,* coronary artery disease; *COPD,* chronic obstructive pulmonary disease; *CVA,* cerebrovascular accident; *DM,* diabetes mellitus; *HTN,* hypertension; *MI,* myocardial infarction; *TIA,* transient ischemic attack.
Excerpted from ASA Physical Status Classification System 2019 of the American Society of Anesthesiologists. A copy of the full text can be obtained from ASA, 1061 American Lane, Schaumburg, IL, 60173-4973 or online at www.asahq.org.

patient's compliance with suggested medication, frequency of physician visits, and most recent visit.

Typical questions include the following: *Do you take medication as prescribed by your physician?* Or, *when was the last time you were examined by your physician?* Other systems have been proposed that would better reflect the increasing number of medically complex patients treated by clinicians as Americans live longer.[35] Regardless of the classification system used, these generalized guidelines need to be individualized for the patient under care.

An alternative means of considering risk assessment is to review the following issues:

- History of allergies
- History of drug interactions, adverse effects
- Presence of prosthetic valves, joints, stents, pacemakers, and so on
- Antibiotics required (prophylactic or therapeutic)
- Hemostasis (normal expected, modification to treatment)
- Patient position in chair
- Infiltration or block anesthesia with or without vasoconstrictor
- Significant equipment concerns (radiographs, ultrasonics, electrosurgery)
- Emergencies (potential for occurrence, preparedness)
- Anxiety (past experiences and management strategy)

A review of these areas provides the clinician with essential background data before initiating treatment.

Common Medical Findings That May Influence Endodontic Treatment Planning

CARDIOVASCULAR DISEASE

Patients with some forms of cardiovascular disease are vulnerable to physical or emotional stress that may be encountered

during dental treatment, including endodontics. Patients may be confused or not understand the specifics of their particular cardiovascular problem.

In those situations, consultation with the patient's physician is mandatory before initiation of endodontic treatment. "For patients with symptoms of unstable angina or those who have had an MI (myocardial infarction) within the past 30 days (major risk category), elective treatment should be postponed."[51] However, one study found "no significant increase in the risk of experiencing a second vascular event after dental visits, including those that involved invasive procedures, in periods up to 180 days after a first recorded ischemic stroke, transient ischemic attack (TIA) or acute MI."[101]

The use of vasoconstrictors in local anesthetics poses potential problems for patients with ischemic heart disease. In these patients, local anesthetics without vasoconstrictors may be used as needed. If a vasoconstrictor is necessary, patients with intermediate clinical risk factors (i.e., a past history of MI without ischemic symptoms) and those taking nonselective beta-blockers can safely be given up to 0.036 mg epinephrine (two cartridges containing 1:100,000 epinephrine) at a single appointment. For patients at higher risk (i.e., those who have had an MI within the past 7 to 30 days and unstable angina), the use of vasoconstrictors should be discussed with the physician.[51]

Vasoconstrictors may interact with some antihypertensive medications and should be used only after consultation with the at-risk patient's physician. Local anesthetic agents with minimal or no vasoconstrictors are usually adequate for nonsurgical endodontic procedures (see also Chapter 6 on pain control).[51] A systematic review of the cardiovascular effects of epinephrine concluded that the increased risk for adverse events among uncontrolled hypertensive patients was low, and the reported adverse events associated with epinephrine use in local anesthetics was minimal.[6] Another review highlighted the advantages of including a vasoconstrictor in the local anesthesia and stated that "pain control was significantly impaired in those patients receiving the local anesthetic without the vasoconstrictor as compared to those patients receiving the local anesthetic with vasoconstrictor."[14]

Some heart conditions make a patient more susceptible to an infection of the heart valves, induced by a bacteremia. That infection, called *infective* or *bacterial endocarditis,* is potentially fatal. In 2008, the American College of Cardiology and American Heart Association (AHA) Task Force on Practice Guidelines published an update on their previous guidelines, which focused on infectious endocarditis. This guideline stated that prophylaxis against infective endocarditis is reasonable for the following patients at highest risk for adverse outcomes from infective endocarditis who undergo dental procedures that involve manipulation of either gingival tissue or the periapical region of teeth or perforation of the oral mucosa: patients with prosthetic cardiac valves or prosthetic material used for cardiac valve repair, patients with previous infective endocarditis, and patients with select congenital heart diseases.[73]

The specific recommendations are summarized in an American Association of Endodontists (AAE) (Chicago, IL) reference guide found online at https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/06/aae_antibiotic-prophylaxis-2017update.pdf. Because the AHA periodically revises its recommended antibiotic prophylactic regimen for dental procedures, it is essential that the clinician stay current concerning this important issue.

There is a low compliance rate among at-risk patients regarding their use of the suggested antibiotic coverage before dental procedures. Therefore, the clinician must question patients concerning their compliance with the prescribed prophylactic antibiotic coverage before endodontic therapy. If a patient has not taken the antibiotic as recommended, it may be administered up to 2 hours after the procedure.[51]

Patients with artificial heart valves are considered susceptible to bacterial endocarditis. Consulting the patient's physician in such cases regarding antibiotic premedication is essential. Some physicians elect to administer parenteral antibiotics in addition to or in place of the oral regimen.

A dentist may be the first to detect elevated blood pressure if he or she routinely evaluates blood pressure before treatment. Furthermore, patients receiving treatment for hypertension may not be controlled adequately because of poor compliance or inappropriate drug therapy. Abnormal blood pressure readings may be the basis for physician referral.

There is a widespread belief among dentists and physicians that oral anticoagulation therapy with drugs such as warfarin (Coumadin, Bristol-Myers Squibb, New York, NY, USA) must be discontinued before dental treatment to prevent serious hemorrhagic complications, especially during and after surgical procedures. Aspirin is a drug commonly used as an anticoagulant on a daily basis without the supervision of a physician. However, clinical studies do not support the routine withdrawal of anticoagulant therapy before dental treatment for patients who are taking such medications.[12,41,51]

When patients report they are receiving an anticoagulant medication, they can benefit from the clinician using the following guidelines:

- Identify the reason why the patient is receiving anticoagulant therapy.
- Know the drug classification, mechanism of action, and its half-life. For instance, 1 to 2 hours for standard heparin, 2 to 4 hours for heparin with low-molecular-weight preparations, and 4 to 5 days for warfarin.[51]
- Assess the potential risk versus benefit of altering the drug regimen.
- Know the laboratory tests used to assess anticoagulation levels. For example, the international normalized ratio (INR) value should be 3.5 or less for patients who are taking warfarin to safely undergo dental or surgical endodontic procedures. However, newer-generation direct thrombin inhibitors, such as dabigatran (Pradaxa) (Pradaxa Boehringer Ingelheim Pharmaceuticals, Ingelheim, Germany), do not require INR monitoring.[51]
- Be familiar with methods used to obtain hemostasis both intraoperatively and postoperatively.
- Be familiar with the potential complications associated with prolonged or uncontrolled bleeding.
- Consult the patient's physician to discuss the proposed dental treatment and to determine the need to alter the anticoagulant regimen.

DIABETES

The Centers for Disease Control and Prevention (CDC, Atlanta, GA) in 2017 reported that an estimated 30.3 million people of all ages, or 9.4% of the US population, had diabetes in 2016. (See National Diabetes Statistics Report available at https://www.cdc.gov/diabetes/pdfs/data/statistics/national-diabetes-statistics-report.pdf.) Diabetes was the seventh leading cause of death in the United States in 2016 (heart disease being the first).[119]

Diabetes mellitus appears to have multiple causes and several mechanisms of pathophysiology.[51] It can be thought of as a combination of diseases that share the key clinical feature of glucose intolerance. Patients with diabetes, even those who are well controlled, require special consideration during endodontic treatment. The patient with well-controlled diabetes who is free of serious complications such as renal disease, hypertension, or coronary atherosclerotic disease is a candidate for endodontic treatment. However, special considerations exist in the presence of acute infections. The non–insulin-controlled patient may require insulin, or the insulin dose of some insulin-dependent patients may have to be increased.[83] When surgery is required, consultation with the patient's physician is advisable in order to consider adjustment of the patient's insulin dosage, antibiotic prophylaxis, and dietary needs during the posttreatment period.

The clinician should ask patients with diabetes, who self-monitor their glucose levels, to bring a glucometer to each visit. If pretreatment glucose levels are below normal fasting range (80 to 120 mg/dL), it may be appropriate to take in a carbohydrate source.[5] A source of glucose (e.g., glucose tablets, orange juice, or soda) should be available if signs of insulin shock (hypoglycemic reaction caused by over-control of glucose levels) occur.[51] Signs and symptoms of hypoglycemia include confusion, tremors, agitation, diaphoresis, and tachycardia.[5] The clinician can avoid a hypoglycemic emergency by taking a complete, accurate history of the time and amount of the patient's insulin and meals. When questions arise concerning the appropriate course to follow, the patient's physician should be contacted or treatment deferred.

Appointments should be scheduled with consideration given to the patient's normal meal and insulin schedule.[83] Usually, a patient with diabetes who is well managed medically and is under good glycemic control without serious complications such as renal disease, hypertension, or coronary atherosclerotic heart disease can receive any indicated dental treatment.[60] However, patients with diabetes who have serious medical complications may need a modified dental treatment plan. For instance, although prophylactic antibiotics are generally not required, it "may be prescribed for a patient with brittle (very difficult to control) diabetes, for whom an invasive procedure is planned but whose oral health is poor and the fasting plasma glucose exceeds 200 mg/dL."[51] Local anesthesia would not be an issue in the presence of well-controlled diabetes, "but for patients with concurrent hypertension or history of recent myocardial infarction, or with a cardiac arrhythmia, the dose of epinephrine should be limited to no more than two cartridges containing 1:100,000 epinephrine."[51]

There is some evidence in support of a relationship between glycemic control and periapical inflammation in diabetic patients.[8] A literature review of human studies also indicated a positive association between diabetes and larger number of periapical lesions.[16] One study determined that although apical periodontitis may be significantly more prevalent in untreated teeth in patients with type 2 diabetes, the disease does not seem to influence the response to root canal treatment.[55] However, other studies suggest that diabetes is associated with a decrease in the success of endodontic treatment in cases with pretreatment periradicular lesions.[30,105] In a prospective study on the impact of systemic diseases on the risk of tooth extraction, it has been also shown that an increased risk of tooth extraction after nonsurgical root canal treatment was significantly associated with diabetes mellitus, hypertension, and coronary heart disease.[113] The nature of such an association has not been fully explained, and may not be a causal relationship.

It is suggested that clinicians discuss the possible negative effect of diabetes on the outcome of endodontic treatment with appropriate patients. Patients with diabetes and other systemic diseases may be best served by referral to an endodontist for treatment planning.

PREGNANCY

Although pregnancy is not a contraindication to endodontics, it may require modification of treatment planning. Protection of the fetus is a primary concern when administration of ionizing radiation or drugs is considered. The most important of safety devices associated with dental radiography is the protective lead apron with thyroid collar. Also important are the use of high-speed film, digital imaging, filtration, and collimation.[93] The American College of Obstetricians and Gynecologists' "Committee Opinion" published in 2013 and reaffirmed in 2017 recommends that clinicians reassure patients that "prevention, diagnosis, and treatment of oral conditions, including dental x-rays (with shielding of the abdomen and thyroid) and local anesthesia (lidocaine with or without epinephrine), are safe during pregnancy" (this document can be accessed at: https://www.acog.org/-/media/project/acog/acogorg/clinical/files/committee-opinion/articles/2013/08/oral-health-care-during-pregnancy-and-through-the-lifespan.pdf). The US Food and Drug Administration also states that "the risk of not having a needed x-ray could be much greater than the risk from the radiation" (refer to https://www.fda.gov/Radiation-EmittingProducts/RadiationEmittingProductsandProcedures/MedicalImaging/MedicalX-Rays/ucm142632.htm).

Over the last three decades, the use of prescription medication in the first trimester has increased by over 60%, and use of four or more medications more than tripled. By the year 2008, approximately 50% of women reported taking at least 1 medication during pregnancy.[61] These data suggest the increasing need for both the patient and the clinician to be informed about the indications, contraindications, and safety of consuming drugs during pregnancy. Box 5.2 presents commonly used dental drugs usually compatible with both pregnancy and breast-feeding. It should be noted that most clinicians are used to the old five-alphabet pregnancy categorization of the drugs (A, B, C, D, and X). In 2014, the US Food and Drug Administration

Box 5.2 Partial List of Drugs Usually Compatible With Both Pregnancy and Breast-Feeding

- Local anesthetics including lidocaine, etidocaine, and prilocaine
- Many antibiotics including penicillins, clindamycin, and azithromycin
- Acetaminophen
- Acyclovir
- Prednisone
- Antifungals including fluconazole and nystatin

changed its Pregnancy and Lactation Labeling. The new labels include "a summary of the risks of using the drug during pregnancy and lactation, a discussion of the data supporting that summary, and relevant information to help health care providers make prescription decisions and counsel women about the use of drugs during pregnancy and lactation" (see https://www.fda.gov/media/100406/download).

One of the areas of concern in treating pregnant patients is to select the safest type of analgesics.

The analgesic of choice during pregnancy had been acetaminophen.[82] However, a link between acetaminophen and childhood asthma has been suggested. Research has found that "the use of acetaminophen in middle to late but not early pregnancy may be related to respiratory symptoms in the first year of life."[78] This finding, although not completely validated, and questioned due to possible confounding factors,[4] should be discussed with pregnant patients when an analgesic is being considered. In a 2016 advisory, the National Health Service of Great Britain advised pregnant patients not to take ibuprofen from the 30th week of pregnancy onwards. They stated that at that stage of pregnancy there is an association with increased risks of complications including heart problems. They went on to advise that before 30 weeks of pregnancy, patients should avoid taking ibuprofen, as it may lead to increased risks of complications including miscarriage (see https://www.nhs.uk/common-health-questions/pregnancy/can-i-take-ibuprofen-when-i-am-pregnant/). Aspirin and NSAIDs also convey risks for constriction of the ductus arteriosus, as well as for postpartum hemorrhage and delayed labor.[82] In a recent large epidemiology study done on 29,078 case and 10,962 control mothers from 1997 to 2011, the authors found that "compared to periconceptional use of acetaminophen, selected birth defects occurred more frequently among infants of women using NSAIDs and/or opioids." However, the authors were not able to definitely determine whether these risks were related to the drugs or to indications for treatment.[37] Due to ongoing research in this area, the clinician is strongly advised to routinely review current information concerning the use of medications before prescribing them for pregnant patients. Other than the CDC, one of the resources for both the clinician and the patient is the following website, a service of the nonprofit Organization of Teratology Information Specialists (OTIS), which is an excellent resource for evidence-based information on the safety of medications and other exposures during pregnancy and while breast-feeding: https://mothertobaby.org.

Based on US Food and Drug Administration pregnancy risk factor definitions,[35] local anesthetics administered with epinephrine generally are considered safe for use during pregnancy. Lidocaine is one of the safest local anesthetics, therefore delaying dental care during pregnancy is not advisable; however, the dose and type of the local anesthetic must be carefully determined for women who have contracted medical conditions that can induce serious pregnancy-related complications.[43] Few anxiolytics are considered safe to use during pregnancy. However, a single, short-term exposure to nitrous oxide–oxygen (N_2O-O_2) for less than 35 minutes is not thought to be associated with any human fetal anomalies, including low birth weight.[51] If a need exists for antibiotic therapy, penicillins, cephalosporins, and macrolides are considered first-line agents.

A major concern is that a drug may cross the placenta and be toxic or teratogenic to the fetus. In addition, any drug that is a respiratory depressant can cause maternal hypoxia, resulting in fetal hypoxia, injury, or death. Ideally, no drug should be administered during pregnancy, especially during the first trimester. If a specific situation makes adherence to this rule difficult, then the clinician should review the appropriate current literature and discuss the case with the physician and patient.[13,58,74]

Further considerations exist during the postpartum period if the mother breast-feeds her infant. A clinician should consult the responsible physician before using any medications for the nursing mother. Alternative considerations include using minimal dosages of drugs, having the mother bank her milk before treatment, having her feed the child before treatment, or suggesting the use of a formula for the infant until the drug regimen is completed. Limited data are available on drug dosages and the effects on breast milk.[51]

In terms of treatment planning, elective dental care is best avoided during the first trimester because of the potential vulnerability of the fetus. The second trimester is the safest period in which to provide routine dental care. Complex surgical procedures are best postponed until after delivery.

MALIGNANCY

Some malignancies may metastasize to the jaws and mimic endodontic pathosis, whereas others can be primary lesions (Fig. 5.2). The most common malignancies metastasize to the jaws are breast, lung, thyroid, and prostate.[56] A panoramic radiograph and a cone-beam computed tomographic image are useful in providing an overall view of all dental structures. When a clinician begins an endodontic procedure on a tooth with a well-defined apical radiolucency, it might be assumed to result from a nonvital pulp. However, pulp sensibility testing is essential to confirm a lack of pulp vitality. A vital response in such cases indicates a nonodontogenic lesion.

Careful examination of pretreatment radiographs from different angulations is important because lesions of endodontic origin would not be shifted away from the radiographic apex in the various images. Alternative methods, such as CBCT, may provide important diagnostic information (see Chapter 2 on radiographic interpretation).

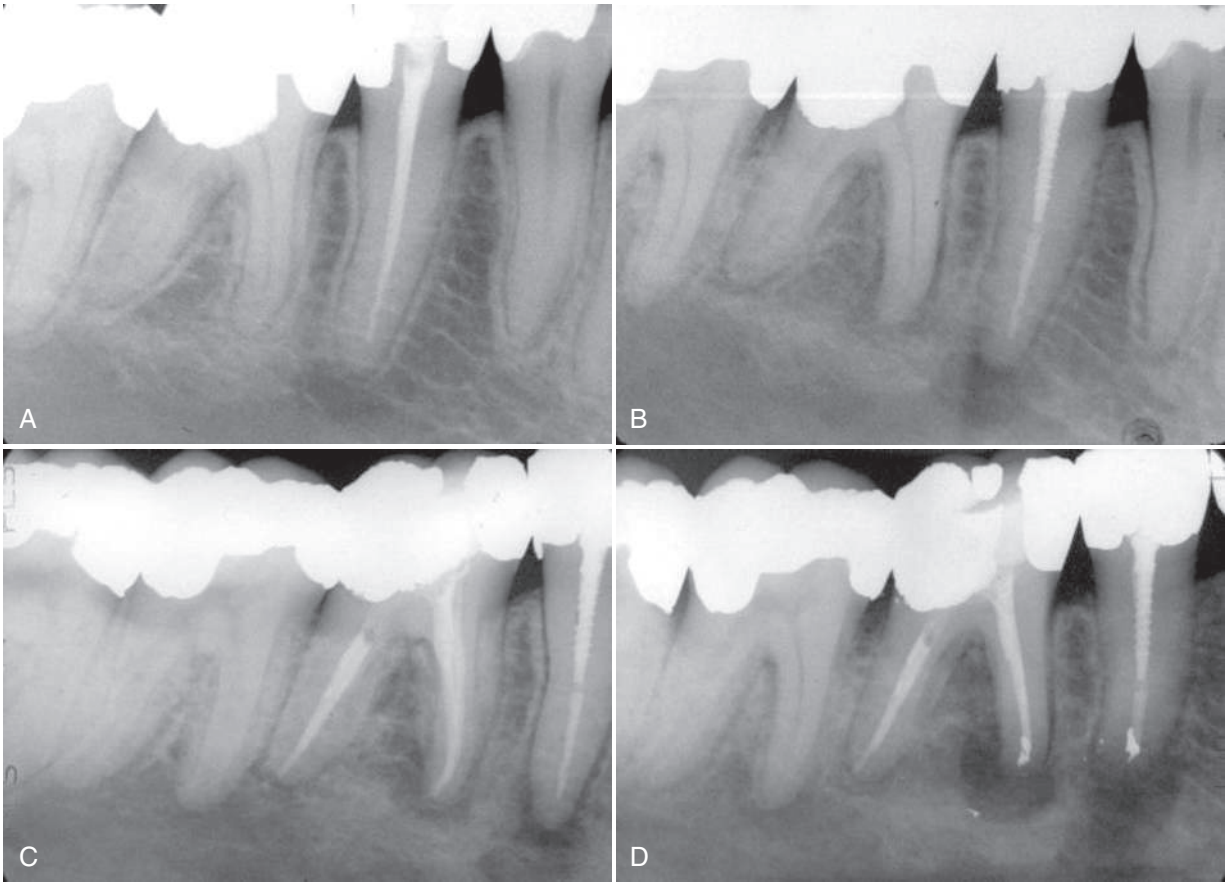


Fig. 5.2 A, Periapical view of tooth #29 after endodontic treatment by a general dental clinician. The diagnosis was irreversible pulpitis. **B,** Patient was referred to an endodontist 4 months later to evaluate radiolucencies of teeth #29 and #30. Symptoms indicated irreversible pulpitis of tooth #30, with concurrent lower right lip and chin paresthesia. Past medical history revealed breast cancer in remission. **C,** Nonsurgical endodontics was performed on tooth #30. Immediate referral was made to an oncologist/oral surgeon for biopsy to rule out nonodontogenic origin of symptoms. **D,** Surgical posttreatment radiograph of teeth #29 and #30. The biopsy report confirmed metastatic breast cancer. (Courtesy Dr. R. Sadowsky, Dr. L. Adamo, and Dr. J. Burkes.)

A definitive diagnosis of periradicular lesions can be made only after biopsy. When a discrepancy exists between the initial diagnosis and clinical findings, consultation with an endodontist is advisable.

Patients undergoing chemotherapy or radiation to the head and neck may have impaired healing responses.[51] Treatment should be initiated only after the patient's physician has been consulted. A dialogue among the dentist, physician, and patient is required prior to determining whether a tooth or teeth should be extracted or endodontically treated prior to radiation.

The effect of the external beam of radiation therapy on normal bone is to decrease the number of osteocytes, osteoblasts, and endothelial cells, thus decreasing blood flow. Pulps may become necrotic from this impaired condition.[51] Toxic reactions during and after radiation and chemotherapy are directly proportional to the amount of radiation or dosage of cytotoxic drug to which the tissues are exposed. Delayed toxicities can occur several months to years after radiation therapy.

Oral infections and any potential problems should be addressed before initiating radiation. It is advised that symptomatic nonvital teeth be endodontically treated at least 1 week before initiating radiation or chemotherapy, whereas treatment of asymptomatic nonvital teeth may be delayed.[51] The outcome of endodontic treatment should be evaluated within the framework of the toxic results of radiation and drug therapy. The white blood cell (WBC) count and platelet status of a patient undergoing chemotherapy should also be reviewed before endodontic treatment. In general, routine dental procedures can be performed if the granulocyte count is greater than 2000/mm^3 and the platelet count is greater than 50,000/mm^3. If urgent care is needed and the platelet count is below 50,000/mm^3, consultation with the patient's physician is required.[51]

MEDICATION-RELATED OSTEONECROSIS OF THE JAWS

Bisphosphonates offer great benefits to patients at risk of bone metastases and in the prevention and treatment of osteoporosis, although this and other drugs (e.g., denosumab) are associated with a rare occurrence of osteonecrosis.[90]

To distinguish medication-related osteonecrosis of the jaws (MRONJ) from other delayed healing conditions, the following working definition of MRONJ has been adopted by the American Association of Oral and Maxillofacial

Surgeons (AAOMS): Patients may be considered to have MRONJ if all of the following three characteristics are present[90]:

1. Current or previous treatment with an antiresorptive drug such as a bisphosphonate or an antiangiogenic drug such as sunitinib (Sutent) (Pfizer, New York, NY, USA), sorafenib (Nexavar) (Bayer, Leverkusen, Germany), or sirolimus (Rapamune) (Pfizer, New York, NY, USA).
2. Exposed, necrotic bone, or bone that can be probed through an intraoral or extraoral fistula in the maxillofacial region that has persisted for more than 8 weeks.
3. No history of radiation therapy to the jaws.

A patient's risk of developing osteonecrosis of the jaw while receiving oral bisphosphonates appears to be low, but there are factors known to increase the risk for MRONJ (Box 5.3). According to aforementioned position statements of the AAOMS, such risks include a history of taking intravenous (IV) bisphosphonates, as opposed to intraoral (IO), taking bisphosphonates for treatment of cancer (high risk), corticosteroid therapy, history of tobacco use, duration of bisphosphonate therapy (>2 to 3 years), a history of an oral operative procedure (i.e., dentoalveolar surgery and extraction), anatomical factors (mandible higher risk than maxilla), concomitant oral diseases (i.e., periodontal disease or periapical pathology), gender (female), age, and genetic predispositions.

Treatment outcomes of MRONJ are unpredictable, and prevention strategies are extremely important. Management of high-risk patients might include nonsurgical endodontic treatment of teeth that otherwise would be extracted. The combination of orthodontic extrusion and bloodless extraction—exfoliation of the extruded roots after their movement—has also been suggested with the aim of minimizing trauma and enhancing the health of the surrounding tissues in patients at risk of developing MRONJ or when a patient refuses to undergo conventional tooth extraction.[102]

For patients at higher risk of MRONJ, surgical procedures such as extractions, endodontic surgery, or placement of dental implants should be avoided. (See https://www.aaoms.org/docs/govt_affairs/advocacy_white_papers/mronj_position_paper.pdf.) Sound oral hygiene and regular dental care may be the best approach to lowering the risk of MRONJ. Patients taking bisphosphonates and undergoing endodontic therapy should sign an informed consent form, inclusive of the risks, benefits, and alternative treatment plans. The following recommendations have been suggested to reduce the risk of MRONJ associated with endodontic treatment[62]:

- Apply a 1-minute mouth rinse with chlorhexidine prior to the start of the treatment with the aim of lowering the bacterial load of the oral cavity.
- Avoid the use of anesthetic agents with vasoconstrictors in order to prevent impairment of tissue vascularization.
- Work under aseptic conditions, including removing of all caries and placement of rubber dam prior to intracanal procedures.
- Avoid damage to the gingival tissues during the placement of rubber dam.
- Avoid maintaining patency of the apical foramen to prevent bacteremia.
- Use techniques that reduce the risk of overfilling and overextension.

Aggressive use of systemic antibiotics is indicated in the presence of an infection in a patient taking bisphosphonates.[51] Discontinuing bisphosphonate therapy may not eliminate any risk of developing MRONJ.[54,57,59] Some clinicians have proposed use of the CTX (C-terminal telopeptide of type I collagen α_1 chain) test (Quest Diagnostics, Madison, NJ) for assessing the risk of developing bone osteonecrosis (BON). For patients who have developed MRONJ, close coordination with an oral maxillofacial surgeon or oncologist is highly recommended.

The AAE updated guideline published on 2018 titled: Endodontic Implications of MRONJ, states: "absent data, the committee considers the risk for ONJ after dental implant placement and endodontic or periodontic procedures that require exposure and manipulation of the bone as comparable to the risk associated with tooth extraction." (See https://www.aae.org/specialty/wp-content/uploads/sites/2/2018/07/AAE_MedRelated_ONJ.pdf.) An astute awareness of the potential risk of MRONJ in patients receiving bisphosphonate therapy is critical. Increased attentiveness to the prevention, recognition, and management of MRONJ will allow the clinician to make the best treatment decisions. Our knowledge of MRONJ is developing rapidly, and it is essential that the clinician monitor the literature for changes in treatment protocols.[54,57,59]

Box 5.3 Risk Factors for Development of Bisphosphonate-Associated Osteonecrosis

- History of taking bisphosphonates for more than 2–3 years, especially with intravenous therapy
- History of cancer, osteoporosis, or Paget disease
- History of traumatic dental procedure
- Patient more than 65 years of age
- History of periodontitis
- History of chronic corticosteroid use
- History of smoking
- History of diabetes

HUMAN IMMUNODEFICIENCY VIRUS AND ACQUIRED IMMUNODEFICIENCY SYNDROME

From 1987 through 1994, human immunodeficiency virus (HIV) disease mortality increased and reached a plateau in 1995. Subsequently, the mortality rate for this disease decreased an average of 33% per year from 1995 through 1998, and 6.2% per year from 1999 through 2016.[119] This dramatic improvement seems to be due to the use of a combination of highly active antiretroviral therapy (HAART) and improved preventive strategies.[51]

It is important, when treating patients with acquired immunodeficiency syndrome (AIDS), that the clinician understand the patient's level of immunosuppression, drug therapies, and the potential for opportunistic infections. Although the effect of HIV infection on long-term prognosis of endodontic therapy is unknown, it has been demonstrated that clinicians may not have to alter their short-term expectations for periapical healing in patients infected with HIV.[80] The clinical team must also minimize

the possibility of transmission of HIV from an infected patient, and this is accomplished by adherence to universal precautions. (See Universal Precautions for Prevention of Transmission of HIV and Other Bloodborne Infections, available at https://www.cdc.gov/niosh/topics/bbp/universal.html.)

Although saliva is not the main route for transmission of HIV, the virus has been found in saliva and its transmission through saliva has been reported.[32] Infected blood can transmit HIV, and during some procedures it may become mixed with saliva. Latex gloves and eye protection are essential for the clinician and staff. HIV can be transmitted by needlestick or via an instrument wound, but the frequency of such transmission is low, especially with small-gauge needles. Nevertheless, "Patients at high risk for AIDS and those in whom AIDS or HIV has been diagnosed should be treated in a manner identical to that for any other patient—that is, with standard precautions."[51]

A vital aspect of treatment planning for the patient with HIV/AIDS is to determine the current $CD4^+$ lymphocyte count and level of immunosuppression. In general, patients having a $CD4^+$ cell count exceeding 350 cells/mm^3 may receive all indicated dental treatments. Patients with a $CD4^+$ cell count of less than 200 cells/mm^3 or severe neutropenia (neutrophil count lower than 500/μL) will have increased susceptibility to opportunistic infections and may be effectively medicated with prophylactic drugs. WBC and differential counts, as well as a platelet count, should be ordered before any surgical procedure is undertaken. Patients with severe thrombocytopenia may require special measures (platelet replacement) before surgical procedures. Care in prescribing medications must also be exercised with any medications after which the patient may experience adverse drug effects, including allergic reactions, toxic drug reactions, hepatotoxicity, immunosuppression, anemia, serious drug interactions, and other potential problems. The practitioner should also be aware of oral manifestations of the disease as far as it concerns diagnosis and treatment planning. For instance, candidiasis of the oral mucosa, Kaposi sarcoma, hairy leukoplakia of the lateral borders of the tongue, herpes simplex virus (HSV), herpes zoster, recurrent aphthous ulcerations, linear gingival erythema, necrotizing ulcerative periodontitis, necrotizing stomatitis, oral warts, facial palsy, trigeminal neuropathy, salivary gland enlargement, xerostomia, and melanotic pigmentation are all reported to be associated with HIV infection. It is essential that consultation with the patient's physician occurs before performing surgical procedures or initiating complex treatment plans.[51,94]

PROSTHETIC IMPLANTS

Patients with prosthetic implants are frequently treated in dental practices. The question concerning the need for antibiotic prophylaxis to prevent infection of the prosthesis has been debated for many years. A brief review of the subject should of interest to clinicians.

A statement was issued jointly in 2003 by the American Dental Association (ADA; Chicago, IL) and the American Academy of Orthopaedic Surgeons (AAOS; Rosemont, IL) in an attempt to clarify the issue.[30] The statement concluded that scientific evidence does not support the need for antibiotic prophylaxis for dental procedures to prevent prosthetic joint infections. It should be noted that although endodontics has been shown to be a possible cause of bacteremia,[21,42] the risk is minimal in comparison with extractions, periodontal surgery, scaling, and prophylaxis.[75] Nevertheless, there had been numerous conflicting opinions about the need to premedicate such patients that were raised over the next 15 years after the publication of the original statement in 2003, such as the 2009 statement of the AAOS entitled Antibiotic Prophylaxis for Bacteremia in Patients with Joint Replacements, which recommended that clinicians consider antibiotic prophylaxis for all total joint replacement patients prior to any invasive procedure that may cause bacteremia. (See American Academy of Orthopaedic Surgeons: AAOS releases new statement on antibiotics after arthroplasty, www.aaos.org/news/aaosnow/may09/cover2.asp.) In 2012, an evidenced-based guideline was published that included recommendations of the AAOS-ADA clinical practice guideline for Prevention of Orthopaedic Implant Infection in Patients Undergoing Dental Procedures. This guideline stated that there is limited evidence for discontinuing the practice of routinely prescribing prophylactic antibiotics for patients with hip and knee prosthetic joint implants undergoing dental procedures. (See American Academy of Orthopaedic Surgeons: Prevention of orthopaedic implant infection in patients undergoing dental procedures, available at www.aaos.org/research/guidelines/PUDP/PUDP_guideline.pdf.) In 2015, the *Journal of the American Dental Association* published an article based on the review of prior research stating that "for patients with prosthetic joint implants, prophylactic antibiotics are not recommended prior to dental procedures to prevent prosthetic joint infection. The practitioner and patient should consider possible clinical circumstances that may suggest the presence of a significant medical risk in providing dental care without antibiotic prophylaxis, as well as the known risks of frequent or widespread antibiotic use."[103] The 2017 update of the AAE's Antibiotic Prophylaxis Reference Guide also reflect this position. (See https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/06/aae_antibiotic-prophylaxis-2017update.pdf.)

Consultation with the patient's physician on a case-by-case basis is advisable to assess the need for prophylaxis.

BEHAVIORAL AND PSYCHIATRIC DISORDERS

Stress reduction is an important factor in the treatment of patients with behavioral and psychiatric disorders. Sensitivity to the patient's needs must be part of the dental team's approach. Significant drug interactions and side effects are associated with tricyclic antidepressants, monoamine oxidase inhibitors, and antianxiety medications.[51] Consultation with physicians in such cases is essential before using sedatives, hypnotics, antihistamines, or opioids.

PSYCHOSOCIAL EVALUATION

The initial visit, during which medical and dental histories are collected, provides an opportunity to consider the patient's psychosocial status and how they value their oral health. Although some patients may want to maintain a tooth with a questionable prognosis, others may lack the ability to comprehend the potential risks and benefits. It

would be a mistake to lead patients beyond what they can understand and appreciate. Furthermore, patients should not be allowed to dictate a treatment plan that has a poor prognosis.

The clinician should also assess the patient's level of anxiety as an important part of preparation for the procedure to follow. It is reasonable to assume that most patients are anxious to some degree, especially when they are about to undergo endodontic treatment. A conversation describing the procedure and what the patient can expect is an important part of an anxiety-reduction protocol.

A study evaluated the most common pathways of fear and anxiety in patients who previously had root canal treatment or were planning to have that procedure. Cognitive conditioning and parental pathways seemed to be the primary cause ($P < .05$) of fear and anxiety with root canal treatment. Females were significantly more likely to be influenced by indirect conditioned experiences such as informative, parental, verbal threat, and vicarious pathways. It was concluded that the origin of patients' fears requires more attention to the importance of treating endodontic-related fear and anxiety. More knowledge about these factors would enable dentists to better manage patients in a multicultural society.[17] More information will be provided in the Anxiety segment of this chapter.

Development of the Endodontic Treatment Plan

The long-term strategic value of a tooth should be considered before presenting an endodontic treatment plan. For example, a restorative plan could be endangered by including a tooth with a questionable endodontic prognosis. Currently, endodontic treatment options exist that were not previously supported by significant levels of evidence. The options include vital pulp therapy, regenerative procedures, and intentional replantation. Although some decisions may be straightforward, considering alternative treatment options can be challenging, as the clinician weighs multiple factors that will play a role in determining the ultimate success or failure of the case. For example, if initial endodontic nonsurgical or surgical procedure has failed, the most common alternative has been an implant or a fixed or removable prosthesis. However, there is increasing evidence that intentional replantation is a cost-effective procedure that may be accomplished, in select cases, with a good prognosis. It is important to note that even if replantation fails, a dental implant remains as an additional option.

Referral of the patient to a specialist should be considered when the complexity of a procedure is beyond the ability of a clinician. The complexity can be due to restorative factors, unusual anatomy, and calcification of the chamber and/or the canal. Other factors that affect endodontic prognosis include periodontal, restorative, and the patient's systemic health.

Endodontic Prognosis

Prognostic studies have identified a number of preoperative factors affecting the outcome of primary endodontic treatment. In a systematic review, it was determined that the absence of periapical radiolucency improves the outcome of root canal treatment significantly. The same study also showed that tooth vitality does not have an impact on the endodontic prognosis, as long as the periapex is healthy.[72] Studies have shown that the size of the radiolucency may also affect the outcome of endodontic treatment.[72,87] It was found that the existence of sinus tract, narrow but deep periodontal probing depth, pain, and discharging sinus have a significant effect on the outcome of nonsurgical root canal treatment.[69] Preoperative pain is not only one of the most important predictors of postoperative pain,[66] it also has an impact on tooth survival after endodontic treatment.[68]

These findings suggest that all preexisting signs and symptoms that may affect the prognosis of treatment, along with the prognostic factors associated with other disciplines—which will be discussed in the next segment—should be taken into consideration when developing a treatment plan. It is of utmost importance that the prognosis and risks and benefits of the treatment be relayed to the patient before the initiation of the treatment as well.

There is a general belief that the prognoses for retreatment cases are poorer than for primary treatment, but this is not universally supported. In a systematic review it was suggested that the outcomes of retreatment cases should be similar to treatment cases as long as access to the apical infection can be reestablished.[67] However, there is some evidence indicating that the incidence of postoperative pain and flare-up is higher in retreatment cases in comparison to treatments.[36] The presence of preoperative periapical lesion, apical extent of root filling, and quality of coronal restoration has been proved to significantly affect the outcome of retreatment cases.[67]

Retreatment cases offer a particular set of challenges to the clinician (Figs. 5.3 and 5.4), and this topic is covered extensively in Chapter 10. Important questions to be considered before retreatment include the following:

- Why did the treatment fail?
- Can the point of bacterial entry to the canal space be identified?



Fig. 5.3 Incision and drainage should be performed on this fluctuant swelling *(arrow)* in conjunction with canal instrumentation.

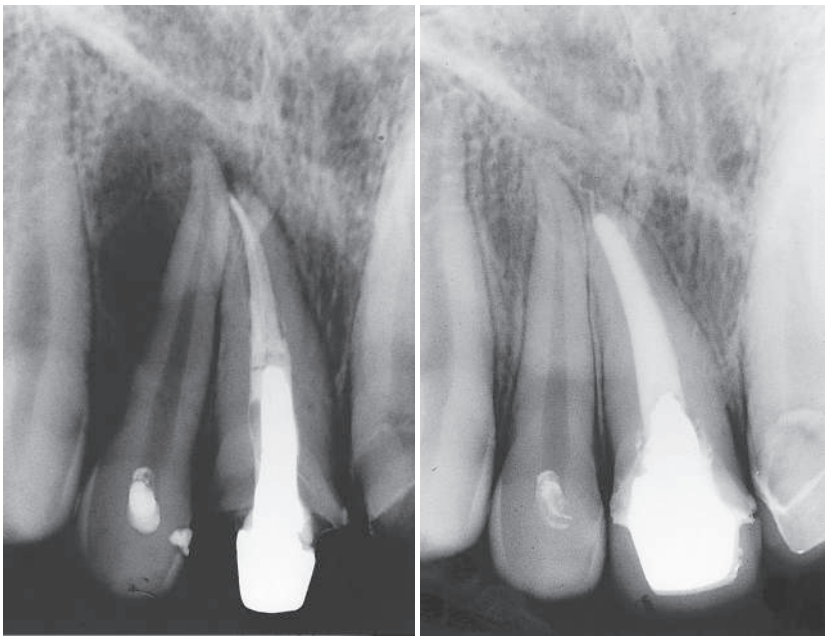


Fig. 5.4 Two years after endodontic therapy of tooth #8, the patient returned with pain and swelling. A clinician mistakenly began endodontic access on tooth #7, without confirming the apparent radiographic diagnosis by sensibility testing. Tooth #7 was vital, and tooth #8 was successfully retreated after removal of the post. (Courtesy Dr. Leon Schertzer.)

- Are prior radiographs available for review?
- Is there an obvious endodontic procedural problem that can be corrected?
- Is the canal system readily accessible for reentry?
- There are additional factors (i.e., a significant periodontal defect or being under immunotherapy) that may have contributed to the failure.
- Is the tooth critical to the treatment plan? (Keeping in mind that including a tooth with a questionable endodontic or periodontal prognosis may endanger an entire prosthesis.)
- Does the patient understand the prognosis for the tooth and want to attempt retreatment? This question and its answer should be documented in the patient's record. The dentist could enter in the patient's record the following statement: "All reasonable options including extraction were presented to the patient and advantages and disadvantages of each were explained."

A retreatment plan should be developed after the clinician has determined the cause of failure and weighed other factors that may affect the prognosis (e.g., root fracture, defective restoration) (Figs. 5.5 to 5.8). Retreatment cases may require surgical endodontics in combination with nonsurgical retreatment. Referral to a specialist is often helpful when planning treatment for complex cases. If retreatment (with or without surgery) of a tooth with a new restoration is being considered, it should be weighed against the possibility of intentional replantation or an implant. Many variables must be considered before a reasonable conclusion can be reached.

SINGLE-VISIT VERSUS MULTIPLE-VISIT TREATMENT

Treatment planning includes considering a multiple or single visit approach. The number of roots, canal anatomy, time available, and the clinician's skills are among the factors to be considered. Severity of the patient's pretreatment symptoms is another important consideration. For example, a patient in severe pain, with or without swelling, is not a prime candidate for single visit endodontics. The initial goal for the patient in severe pain should be directed at alleviating pain, with filling of the canal deferred until the symptoms have been brought under control. The clinician's judgment of what the patient can comfortably tolerate (regarding duration of the visit) is made on a case-by-case basis. The second visit allows the clinician to determine the effect of the initial treatment and then complete instrumentation and filling if the tooth is no longer symptomatic.

Numerous studies have compared single and multiple visit cases and reached different conclusions about the outcomes of each approach. For example, some studies have reported less posttreatment pain in single-visit cases,[52,84,104] whereas a systematic review found that the incidence of postobturation discomfort was similar in the single- and multiple-visit approaches.[29] In another systematic review, it was concluded that there was a lack of compelling evidence, indicating a significantly different prevalence of posttreatment pain/flare-up with either single- or multiple-visit root canal treatment.[92] Differences in research methodology explain the conflict in these findings.

Regarding the healing rate of single- and multiple-visit cases, systematic reviews have found that there is no detectable difference in the effectiveness of root canal treatment in terms of radiologic success between single and multiple visits,[29] even in cases of infected teeth.[104]

Teeth with nonvital pulps and apical periodontitis pose a microbiologic problem. It is important to avoid pushing bacterial debris into the periapical tissues. Agreement is lacking concerning the appropriateness of single-visit endodontics for treating these patients. Some have postulated that the

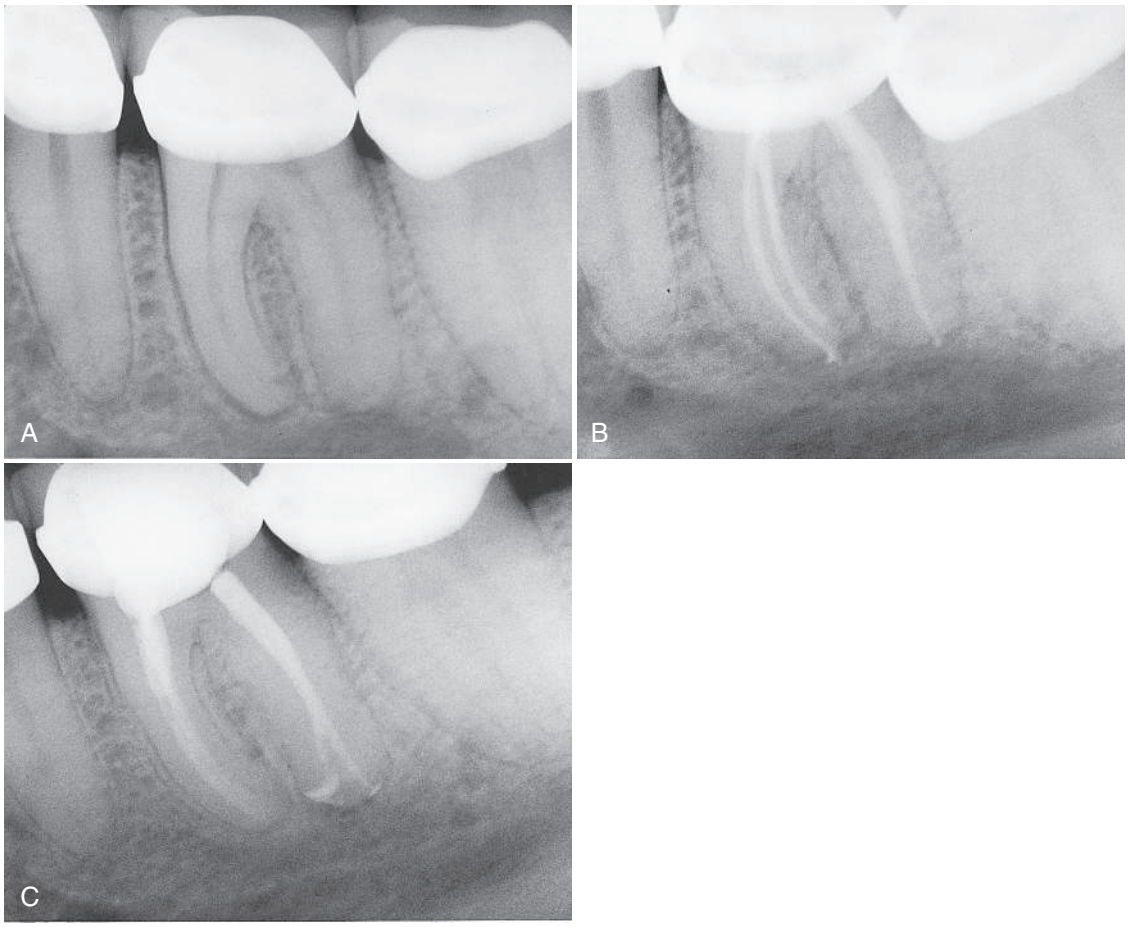


Fig. 5.5 Many years after endodontic treatment of tooth #19, the patient returned with a chief complaint of pain and an inability to chew with the tooth. Despite the radiographic appearance of excellent endodontic treatment, the tooth was retreated and the patient's pain disappeared. Note the unusual distal root anatomy, which was not apparent during the initial procedure. **A,** Initial radiograph. **B,** Completion of initial endodontic therapy. **C,** Retreatment.

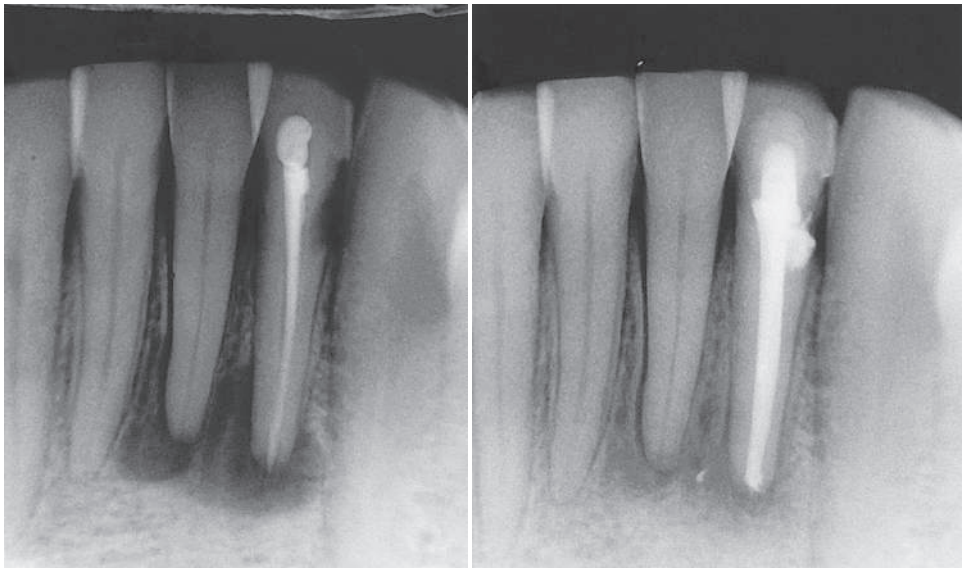


Fig. 5.6 Initial radiograph was misleading and implicated tooth #23 and tooth #24. Pulp testing indicated a vital pulp in tooth #24, and it was not treated. Retreatment of #23 resulted in healing of the periradicular lesion. (Courtesy Dr. Leon Schertzer.)

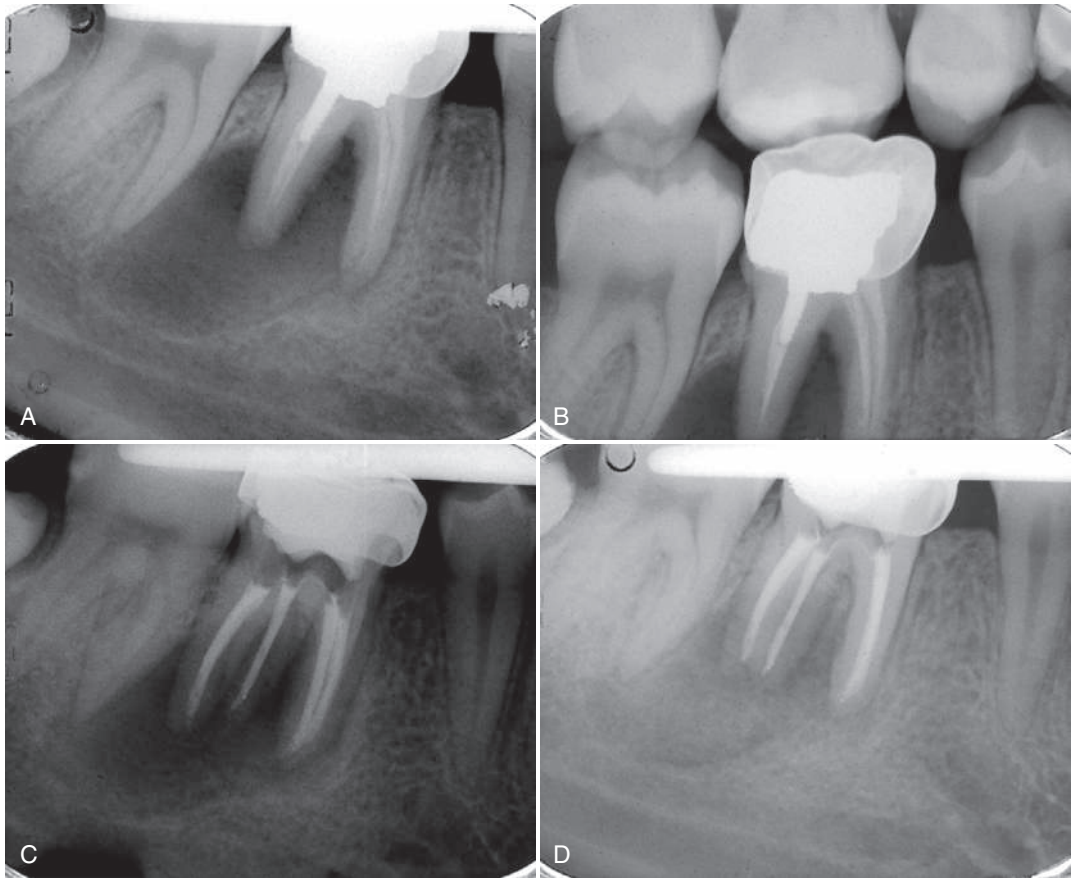


Fig. 5.7 Nonsurgical retreatment of tooth #30. An additional root was located and treated. **A,** Note inadequate endodontic treatment and large periapical lesion. **B,** Bitewing radiograph. **C,** Retreatment after post removal. **D,** Eighteen-month recall radiograph indicates periapical healing.

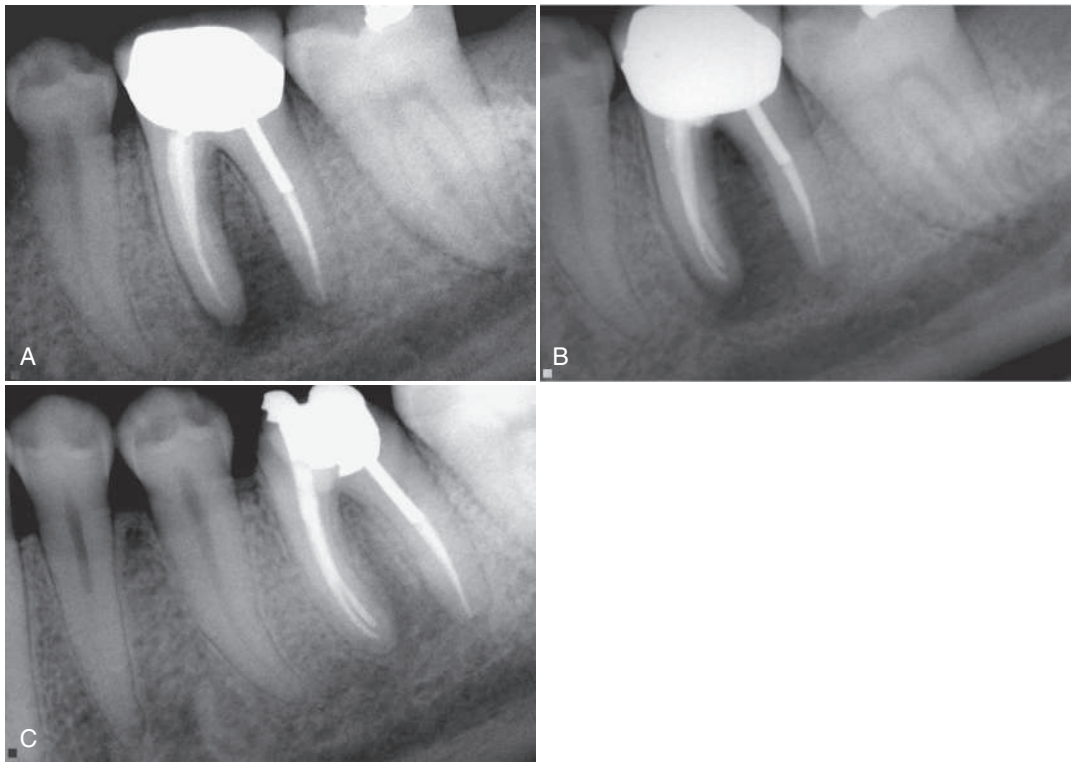


Fig. 5.8 Despite several exacerbations during endodontic retreatment, this case responded rapidly after the completion of therapy. Radiographic evaluation indicates a good periapical response after only 4 months. **A,** Radiograph taken 2 years after initial root canal therapy. **B,** Retreatment of the mesial root. **C,** Radiograph taken 4 months following retreatment.

intervisit use of an antimicrobial dressing is essential to thoroughly disinfect the root canal system.[99,100,110] In contrast, other researchers have found no statistically significant difference in success when using the single-visit or multiple-visit approach to the nonvital tooth with apical periodontitis.[29,63,77,79,87,91,116]

Differences in research findings concerning single or multiple visits might be due to variations in research methodology, including sample size, duration of follow-up, and treatment methods.

It is possible that total elimination of bacteria may not be absolutely necessary for healing. Perhaps maximal reduction of bacteria, effective root canal filling, and a timely satisfactory coronal restoration can result in a high level of clinical success. However, regardless of the number of appointments, effective bacteriologic disinfection of the root canal system is critical.

Treatment planning for an endodontic case should be based on biologic considerations. Patients who present with acute symptoms present a different set of biologic and clinical issues than those with an asymptomatic tooth. Swelling associated with an abscess, cellulitis, or presence of a stoma represent signs of pathologic processes. The biologic significance of these conditions should be considered before determining specific goals for each visit.

Developing goals for each visit helps to organize the treatment. For example, an asymptomatic anatomically uncomplicated anterior, premolar, or molar tooth may be a candidate for a single-visit plan, depending on the skills of the practitioner. However, as the complexity of the case increases and/or symptoms become more severe, a multivisit approach becomes more appropriate.

These recommendations have a biologic basis. Biologically, it is not reasonable to partially instrument root canal systems, thereby leaving residual inflamed pulpal remnants or necrotic debris in the canal, because such remnants may cause pain and be susceptible to infection. The clinician would be well advised to begin canal instrumentation only if time permits the extirpation of all pulp tissue and complete debridement of the root canal system.

In most cases, the clinical procedures required to complete endodontic treatment can be accomplished in a single visit. However, that does not mean it is the wiser course of treatment. What can be done and what should be done represent two very different approaches to endodontic treatment. The patient's systemic health, level of anxiety, and symptoms, as well as the complexity of the root canal system, and clinician's experience are all factors that must be considered.

A study concerning the outcome of initial endodontic treatment noted the complexity of treating apical periodontitis.[87] The author commented that treatment of this disease cannot be improved merely by changing treatment techniques. Because apical periodontitis results from interactions between microorganisms, their environment, and the host immune system, only the use of effective modifiers of any of these three factors might significantly improve the outcome of treatment.

SURGICAL ENDODONTICS

Surgical evaluation is particularly valuable in the diagnosis of lesions that may be nonodontogenic. Biopsy is the definitive means of diagnosing osseous pathosis, which may mimic a lesion of endodontic origin. When retreatment is being considered, the clinician must determine whether nonsurgical, surgical, or combined treatment is appropriate. This decision is influenced by the presence of complex restorations, posts, and the radiographic assessment of prior endodontic therapy.

Endodontic surgery is most often performed in an attempt to improve the apical seal and correct failure of nonsurgical therapy. Bacteria are the essential cause of failure. It is important that the clinician determine the bacterial path of ingress. For example, a deficient restoration or recurrent decay will result in microleakage into the root canal space. Unless that issue is addressed, the outcome of apical surgery may not be predictable (Fig. 5.9).

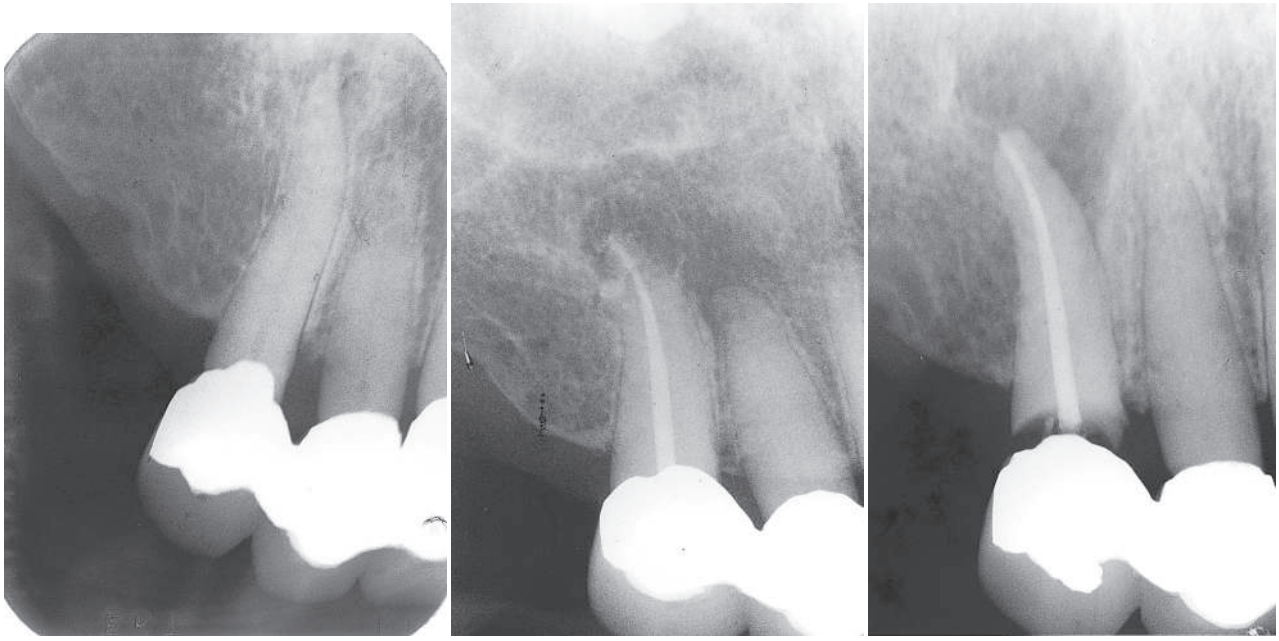


Fig. 5.9 Four years after endodontic therapy, the patient complained of pain and swelling associated with tooth #6. The initial impression was that apical surgery was indicated. However, further radiographs revealed the true cause of the endodontic failure. The initial endodontic access through the crown or caries damaged the coronal seal and recurrent decay followed.

When a deficient restoration is identified, it must be replaced to prevent continued bacterial penetration. Endodontic surgery (see Chapter 11) may also be performed as a primary procedure when there are complications such as calcific metamorphosis. In those cases, by using surgery as primary therapy, an apical seal can be established while preserving the crown of the tooth. The treatment plan for these cases is determined after reviewing multiple radiographs and considering the possibility of completing nonsurgical therapy without destroying an otherwise functional crown or natural tooth. Endodontic surgery without prior nonsurgical therapy should be considered a treatment of last resort and only when nonsurgical treatment is not possible.

Reviewing the best available evidence for alternative treatments is an important aspect of treatment planning for a tooth with failed endodontics. Evidence concerning the healing potential after endodontic surgery is an important consideration in the management of posttreatment disease.[86] Numerous studies have examined the outcome of apical surgery and the results vary considerably.[111,114,121] This variability may reflect actual outcome differences or reflect variations in case selection techniques, recall periods, and methodology.

A prospective study indicated that there is an increased odds ratio for disease persistence for teeth with larger pretreatment lesions and pretreatment root canal filling of adequate length.[114] Another study found that patients presenting with pain at the initial examination before surgery had a significantly lower rate of healing at the 1-year follow-up compared with patients who did not have pain at the initial examination.[111]

It should also be noted that the periodontal condition of the tooth, including the interproximal bone levels and the amount of marginal bone loss, has been shown to significantly affect the long-term prognosis of periapical surgery.[112,115] Moreover, it has been shown that isolated endodontic lesions have much higher success rates at the 1- to 5-year follow-up (95.2%) in comparison with endodontic-periodontal combined lesions (77.5%).[46]

Dramatic changes in surgical technique and materials have occurred. The advent of microscopy, endoscopy, and ultrasonics, as well as improved retrograde filling materials, represents important modifications of surgical technique. An outcome study comparing traditional root-end surgery (TRS) and endodontic microsurgery (EMS) found that the probability of success for EMS was 1.58 times greater than the probability of success for TRS.[97] Another study comparing the EMS techniques with and without the use of higher magnification found that the difference of probability of success between the groups was statistically significant for molars, but no significant difference was found for the premolar or anterior teeth.[96] CBCT has proved to be of value in some surgical cases, producing three-dimensional images of a tooth, pathosis, and adjacent anatomic structures. It is beneficial for localizing the mandibular canal,[48] mental foramen, maxillary sinus, and nasal cavity.[18]

INTENTIONAL REPLANTATION

Intentional replantation is a procedure that has long been thought of as a measure of last resort and rarely considered during treatment planning. Its increasing popularity may be based on evidence indicating that its success rate, when specific guidelines are followed, may approach that of other endodontic procedures and dental implants. When compared to dental implants, the procedure offers a more cost-effective and timely means of maintaining a tooth that would otherwise be lost. In terms of patient-centered outcomes, replantation should be considered prior to extraction of previously endodontically treated teeth. While nonsurgical or surgical endodontic treatment remains the preferred treatment for most patients, evidence is building supporting the concept that replantation of a tooth is a valuable option for some patients. It should be noted that in those cases, where replantation is not successful, a dental implant remains an option.

In 2017, a systematic review compared the survival of teeth intentionally replanted, using a modern technique and cost effectiveness, compared with single-tooth implants. Six studies met the inclusion criteria. Meta-analysis resulted in a survival rate of 89.1%. Compared with a single- tooth implant, intentional replantation was more cost-effective, even when custom post/core and crown were also needed. The authors concluded that based on the meta-analysis, there was a high survival rate for intentional replantation. Although the survival rate for implants was higher, intentional implantation is a more cost-effective treatment modality. The authors concluded that, "When nonsurgical and surgical retreatments of a tooth are not feasible or have a poor prognosis, intentional replantation with a modern technique is a reliable and cost-effective method of treating teeth. The option of intentional replantation should be discussed with patients as an alternative to extraction and implant placement."[53] (For further information please visit Chapter 11.)

REGENERATIVE ENDODONTICS

Regenerative endodontics is a developing area in the treatment of immature teeth with infected root canals that has been described as a "paradigm shift"[22] in the management of these teeth and may result in continued root maturation and apical closure. Traditional approaches using calcium hydroxide for apexification and apical barrier techniques using mineral trioxide aggregate (MTA) have been used in the treatment of immature teeth with pulp necrosis. When using that technique, there is usually no further root development. The roots remain thin and fragile with a high risk of fracture and tooth loss.[81] It has been suggested that regenerative endodontic procedures (REPs) that utilize endogenous stem cells that are introduced in the canal by lacerating the periapical tissues to fill the canal with blood should be used for the treatment of immature teeth with pulp necrosis. Regenerative endodontics has been defined as "biologically based procedures designed to replace damaged structures, including dentin and root structures, as well as cells of the pulp-dentin complex."[22] The AAE recently updated the developed clinical considerations for regenerative procedures; this can be found on the AAE website (https://www.aae.org/specialty/wp-content/uploads/sites/2/2018/06/ConsiderationsForRegEndo_AsOfApril2018.pdf). Most studies that have compared REPs with the traditional approaches of calcium hydroxide apexification and MTA apical barrier techniques have shown

comparable outcomes.[3,44] However, a study that compared traditional and regenerative treatment protocols showed revascularization was associated with significantly greater increases in root length and thickness compared with calcium hydroxide apexification and MTA barrier placement as well as higher overall survival rates.[40]

There are several factors that must be taken into consideration when planning REP treatment for a tooth, such as: (a) remaining tooth structure, (b) tooth location, (c) level of tooth maturity, (d) number of visits, and (e) patient's compliance.

It must be noted that although REP is becoming more common, the traditional apexification techniques still have their own place in the treatment planning process for a premature tooth. For instance, the amount of remaining crown structure is one of the factors to be considered. Placing a post after a REP is usually difficult; therefore, if the tooth has lost most of its crown structure and may need a post in the future, other treatment modalities such as traditional $Ca(OH)_2$ or MTA plug apexification should be considered. Another factor to consider in the treatment planning stage is the location of the tooth. New materials are being introduced on a frequent basis, and each one may or may not be a suitable choice for all teeth. It is of utmost importance for the clinician to refer to the highest level of scientific evidence in treatment planning such cases. For example, MTA is the most common material used in REP; however, one of the common complications with MTA is tooth discoloration, which is a serious esthetic concern with anterior teeth. In these teeth, other materials, such as Biodentine, may be favored.[7] With regards to apical maturity, REP is mostly considered for immature teeth, since this area is rapidly developing, it is important for the clinician to have a broad knowledge on the protocols and biomaterials.

Tooth maturity is another factor to consider. Although it has been suggested that less mature teeth are more suitable for regenerative endodontics,[47] there have been reports of using REP in treating teeth with necrotic pulps and apical periodontitis.[106] Therefore, it is of utmost importance that the clinician discuss all treatment options prior to suggesting the definitive treatment.

The number of visits and patient's compliance are other critical elements in treatment planning for premature teeth. According to the AAE clinical protocol (mentioned earlier), RET usually requires two treatment and few follow-up visits; however, $Ca(OH)_2$ calcification requires multiple observation visits prior to definitive treatment session. It is important to discuss the number of sessions and their frequency with the patient or the parent/guardian in terms of patient's availability and compliance. Patients with fewer complaints may be more suitable for less technique-sensitive treatments or ones with fewer required number of sessions. (For more information refer to Chapter 12.)

Interdisciplinary Treatment Planning

PERIODONTAL CONSIDERATIONS

Complex cases may require a multidimensional approach. For example, extensive periodontal lesions may complicate the prognosis of endodontic treatment. A periodontist and an endodontist might be able to add their insights to the prognosis of treatment and avoid future problems.

A 4-year retrospective study found that attachment loss and periodontal status affected endodontic prognosis of endodontically treated molars.[95] It is crucial that the dental practitioner be aware of the periodontal factors that may influence the prognosis of endodontic treatment, such as root perforations, bone loss, and clinical attachment loss.

When establishing the prognosis of a tooth with an endodontic/periodontal lesion, there are essential factors to be considered. Determination of pulp vitality and the extent of the periodontal defect are central to establishing the prognosis and developing a treatment plan for a tooth with an endodontic/periodontal lesion (see also Chapter 25).

In primary endodontic disease, the pulp is nonvital, whereas in primary periodontal disease, the pulp retains vitality. True combined endodontic-periodontal disease occurs less frequently. The combined lesion is found when the endodontic disease process advances coronally and joins with a periodontal pocket progressing apically. There is significant attachment loss with this type of lesion, and the prognosis is questionable.[85] The radiographic appearance of combined endodontic-periodontal lesions may be similar to that of a vertically fractured tooth. Therapy for true combined lesions requires both endodontic and periodontal therapy. Sequencing of treatment is based on addressing the initial chief complaint.

The prognosis and treatment of each type of endodontic-periodontal disease vary. Primary endodontic disease should be treated solely by endodontic therapy, and the prognosis is usually good (Figs. 5.10 and 5.11). Primary periodontal disease should be treated only by periodontal therapy, and the prognosis varies depending on the severity of the disease and patient's response to treatment.[85]

Pathogenesis of the lesion can be better understood after sensibility testing, periodontal probing, radiographic assessment, and evaluation of dental history. When extensive prostheses are planned, the potential risk of including a tooth with a questionable prognosis must be considered. It is not prudent to incorporate a chronic problem into a new complex prosthesis (Fig. 5.12).

RESTORATIVE AND PROSTHODONTIC CONSIDERATIONS

A satisfactory restoration may be jeopardized by a number of factors. Subosseous root caries (perhaps requiring crown lengthening), poor crown-to-root ratio, and extensive periodontal defects or misalignment of teeth may have serious effects on the final restoration. These problems must be recognized before endodontic treatment is initiated. For complex cases, a restorative treatment plan should be in place before initiating endodontic treatment (see Chapter 23). Some teeth may be endodontically treatable but nonrestorable, or they may represent a potential restorative complication because of a large prosthesis. Reduced coronal tooth structure under a full-coverage restoration makes endodontic access more difficult because of reduced visibility and lack of radiographic information about the anatomy of the chamber. It is not unusual for restorations to be compromised during endodontic access (see Fig. 5.12). Whenever



Fig. 5.10 **A,** Inflamed, edematous interproximal tissue *(arrow)* caused by acute endodontic pathosis. **B,** Soft-tissue healing *(arrow)* 3 days after initiation of endodontic treatment. **C,** Periradicular pathosis. **D,** Completed endodontic therapy. **E,** Periradicular healing at 1-year recall.

possible, restorations should be removed before endodontic treatment.

Full coverage restorations are usually suggested after endodontic treatment. In a systematic review on tooth survival following nonsurgical root canal treatment, four factors were found to be of significance in tooth survival[70]:

- A crown restoration after root canal treatment.
- Tooth having both mesial and distal proximal contacts.
- Tooth not functioning as an abutment for removable or fixed prosthesis.
- Tooth type or specifically nonmolar teeth.

Another systematic review found that the odds for healing of apical periodontitis increase with both adequate root canal treatment and adequate restorative treatment. However, poorer clinical outcomes may be expected with adequate root filling/inadequate coronal restoration and inadequate root filling/adequate coronal restoration, with no significant difference in the odds of healing between these two combinations.[34] These findings suggest that the quality of the coronal restoration is as important as the quality of the root canal treatment. Therefore, to increase the success of the treatment, it is strongly suggested that the clinician discuss the restorative plan of the tooth with both the patient and—if it is a referred patient—with the referring dentists before initiation of treatment.

ENDODONTIC THERAPY OR DENTAL IMPLANT

The successful evolution of dental implants as a predictable replacement for missing teeth has had a positive impact on patient care. A clinician now has an additional possibility to consider when developing a treatment plan for a patient with a missing tooth or teeth. More challenging is the decision concerning whether or not to provide endodontic therapy for a tooth with a questionable prognosis or extract and use a single-tooth implant as a replacement or attempt replantation. Numerous studies have evaluated both the nonsurgical endodontic therapy[20,71,72,88,89,107] and endosseous dental implants.[1,19,38]

It is not possible to compare outcome studies because of variations in research methodologies, follow-up periods, and criteria associated with determining success or failure. A review of outcome studies points to the need for randomized controlled trials with standardized or similar methodologies that could provide a higher level

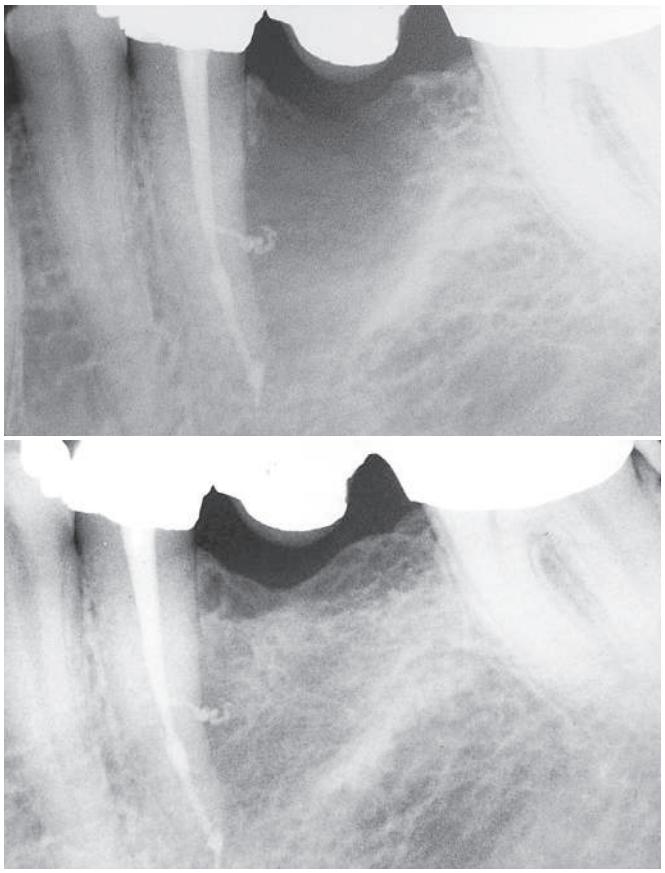


Fig. 5.11 A large, bony defect associated with tooth #20 healed after endodontic therapy. The tooth was nonvital, and no significant periodontal probing depth indicated pulpal disease.



Fig. 5.12 Tooth #30 has a poor prognosis. Periodontal probing reached the apex of the distal root. Extraction is indicated and should be done as soon as possible to prevent further damage to the mesial bone associated with tooth #31. Implant site preservation is another consideration in treatment planning for this case.

of evidence to use in answering important clinical prognostic questions.

A synthesis of available evidence indicates that both primary root canal treatment and single-tooth implants are highly predictable procedures when treatment is appropriately planned and implemented. A study assessed clinical and radiographic success of initial endodontic therapy of 510 teeth over a 4- to 6-year period. It was found that 86% of teeth healed and 95% remained asymptomatic and functional.[20] Another study considered the outcomes of endodontic treatment on 1,462,936 teeth. More than 97% of teeth were retained after 8 years.[88] In a similar study, it was found that overall, 89% of the 4744 teeth were retained in the oral cavity 5 years after the endodontic retreatment.[89] A study concerning the outcome of endodontic therapy on 1312 patients in general practice with a mean follow-up time of 3.9 years found a combined failure rate of 19.1%, concluding that "failure rates for endodontic therapy are higher than previously reported in general practices, according to results of studies based on dental insurance claims data."[9]

The ADA's Council on Scientific Affairs has reported high survival rates for endosseous implants. An evaluation of 10 studies with more than 1400 implants demonstrated survival rates ranging from 94.4% to 99% with a mean survival rate of 96.7%.[1] In a recent systematic review, the authors claimed that with the mean range of 13.4 years follow-up, the survival rate of dental implants were 94.6% and the success rate for half of the cases, which used the same criterion with a mean period of follow-up of 15.7 years, was 89.7%, but a statistical analysis was not performed due to heterogenicity.[108] With such relatively high survival rates reported for endodontics and single-tooth implants, a clinician must consider a multiplicity of factors within the context of the best available evidence.

In a retrospective cross-sectional comparison of initial nonsurgical endodontic treatment and single-tooth implants, it was suggested that restored endodontically treated teeth and single-tooth implant restorations have similar survival rates, although the implant group showed a longer average and median time to obtain function and a higher incidence of posttreatment complications requiring subsequent treatment intervention.[24] Aside from success, the clinician must also consider the success rate of dental implants, which is usually significantly lower than success rates. The term "success" usually involves more complex parameters and criteria and is associated with the health and quality of the implants.[108] Any treatment intervention due to postoperative complications will affect the success of the implant without directly affecting its survival. In a study on long-term survival and success of dental implants, the authors found that "The cumulative complication rate after an observation period of 10 to 16 years was 48.03%, which meant that substantial amounts of chair time were necessary following implant placement."[98] In a recent study of patients who have received both nonsurgical root canal treatment and single-tooth implants, it was concluded that "the number of adjunct and additional treatments, the number of appointments, the elapsed time before the final restoration, the number of prescribed medications, and the cost of the treatment were significantly higher for single tooth implants in comparison with nonsurgical root canal treatment."[109]

A review summarized the best available evidence concerning factors influencing treatment planning involving preservation of a tooth with endodontic therapy or replacement by a single-tooth implant. Factors considered included prosthetic restorability of the natural tooth, quality of bone,

aesthetic concerns, cost-to-benefit ratio, systemic factors, potential for adverse effects, and patient preferences.[38] The authors concluded that "endodontic treatment of teeth represents a feasible, practical and economical way to preserve function in a vast array of cases and that dental implants serve as a good alternative in selected indications in which prognosis is poor."[38]

Aside from treatment outcome, number of sessions, duration of the treatment, and additional adjunct treatments, there are other factors involved in any treatment that the practitioner has to consider when making a treatment planning decision. In a study evaluating the quality of life of endodontically treated versus implant treated patients, the results showed a high rate of satisfaction with both treatment modalities.[31] One study found that when compared to endodontic molar retreatment and fixed partial dentures, implant-supported restoration, despite its high survival rate, has been shown to be the least cost-effective treatment option.[48] Another study found that "endodontically treated natural teeth may provide more effective occlusal contact during masticatory function compared with implant-supported restorations, leading to more efficient mastication."[117]

Another important factor is the patient's health status. Implants require a surgical procedure that may not be possible due to the patient's medical status. One area of concern is diabetes mellitus. However, it has been shown that dental implant osseointegration can be accomplished in these subjects, as long as they have good glycemic control.[39]

It seems clear that patients are best served by retaining their natural dentition as long as the prognosis for long-term retention is positive. It is not reasonable to extract a tooth if endodontics with a good prognosis can be completed. It is also not reasonable for a patient to invest in root canal therapy, a post, and a crown if the prognosis is highly questionable and an implant with a good prognosis can be placed. An important advantage of providing endodontic therapy is to allow rapid return of the patient's compromised dentition to full function and aesthetics. This rapid return is in marked contrast to the use of provisional restorations associated with dental implants while waiting for osseous integration. The challenge is to weigh all pretreatment variables and reach a reasonable conclusion concerning the prognosis for tooth retention or implant placement.

Interestingly, some endodontic advanced education programs are now including implant training in their curricula. This training will enable the endodontist to provide more value to the patient and referring the dental clinician as treatment plans are determined. Such dually trained endodontists will be well positioned to provide endodontic therapy or place an implant as best serves the patient.

Finally, the AAE Position Statement on Implants referring to the Ethics of Clinical Practice states: "Inappropriate treatment, such as: performing endodontic therapy on nonrestorable or periodontally hopeless teeth; or placing single-tooth implants when the natural tooth could predictably be retained, would also be considered unethical. Failure to adhere to these principles not only violates the trust placed in the dental profession, but leaves the dentist vulnerable to litigation" (see https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/06/implantsstatement.pdf).

Other Factors That May Influence Endodontic Case Selection

A variety of factors may complicate proposed endodontic therapy. Calcifications, dilacerations, or resorptive defects may compromise endodontic treatment of a tooth with potentially strategic value (Fig. 5.13).

Diagnosis and treatment of resorptive defects may be challenging at times. Distinguishing external resorption

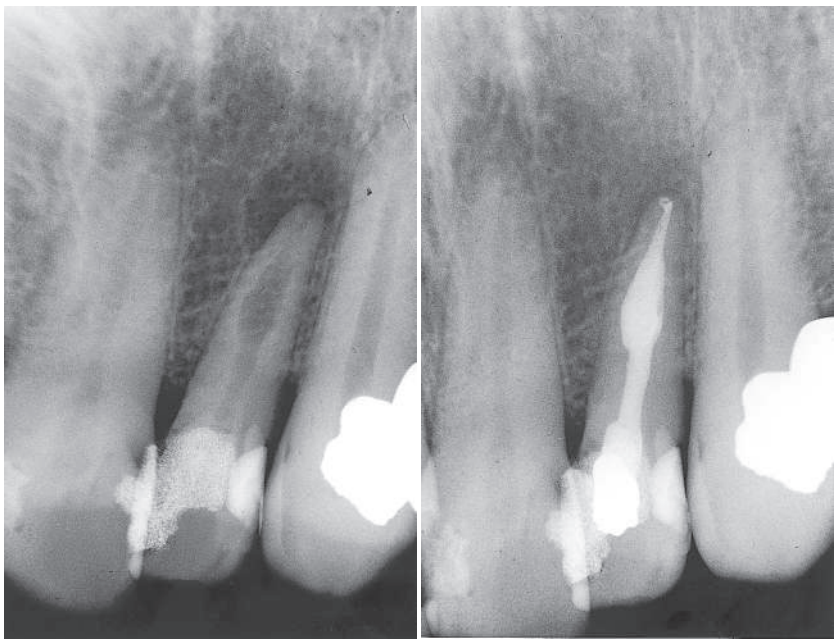


Fig. 5.13 Resorptive defects can be successfully treated. Early intervention, before there is perforation of the root, increases the chance of success. (Courtesy Dr. Leon Schertzer.)

from internal resorption is usually possible with routine radiographs using different angles; but in order to detect the exact location and extension of certain types of resorption, especially external cervical resorption (ECR), the use of the CBCT is paramount for treatment planning purposes.[28,76] Resorptive lesions are usually asymptomatic, and therefore, are often found accidentally when reviewing radiographs. As a result, in many cases, when the lesion is visible on a radiograph, a large portion of the tooth is already affected, making the treatment extremely difficult, if not impossible. In addition, when one tooth in a patient is affected, it is possible that other teeth may also be affected in the same dentition,[33] making the use of CBCT more justified in these cases (Fig. 5.14). Treatment of these lesions usually includes a combination of nonsurgical and surgical endodontic techniques (such as osteotomy and root repair, with or without intentional replantation) (for more information see Chapter 18).

The inability to isolate a tooth is also a problem and may result in bacterial contamination of the root canal system. Extra roots and canals pose a particular anatomic challenge that radiographs do not always reveal (Fig. 5.15). A bitewing radiograph is useful in providing an accurate image of the pulp chambers of posterior teeth.

The clinician should recognize these potential problems and be able to manage and factor them into the decision concerning the tooth's prognosis, including the possibility that the patient should be referred to a specialist.

As mentioned, CBCT is a tool that when used judicially can be helpful in identifying roots that are not visible with conventional radiography. For more information on clinical applications of CBCT, refer to the 2015/2016 AAE and American Academy of Oral and Maxillofacial Radiology (AAOMR) Joint Position Statement (available at https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/06/conebeam-statement.pdf). The statement provides 11 specific recommendations and supporting evidence for when CBCT should be considered, and emphasizes that CBCT should not be routinely used for endodontic diagnosis or screening in the absence of clinical symptoms. Consistent with the principles of ALARA—keeping patient radiation doses "As Low as Reasonably Achievable"—the statement notes that the patient's history and clinical examination must justify the use of CBCT. (Refer to https://f3f142zs0k2w1kg84k5p9i1o-wpengine.netdna-ssl.com/specialty/wp-content/uploads/sites/2/2017/06/conebeamstatement.pdf.)

The revised position statement concludes with the following advisory: "CBCT is an emerging technology that is revolutionizing the approach to endodontic care of dental patients. The guidance in this statement is not intended to substitute for a clinician's independent judgment in patient care. The use of limited field of view (FOV) CBCT should be considered on a case-by-case basis, with due consideration given to the risks and benefits of exposing the patient to ionizing radiation, the patient's history, clinical findings and pre- existing radiographs so that superior treatment can be provided to the general public in need of endodontic care."

Another consideration is the stage of maturity of the tooth. Primary and immature permanent teeth may have a pulpal pathosis caused by caries or trauma; preserving these young teeth is essential. Premature loss of an anterior tooth can lead to malocclusion, predispose the patient to tongue habits, impair aesthetics, and damage the patient's self-esteem. (See Chapter 21 on dental traumatic injuries.)

Some clinicians use a simple formula for determining which endodontic cases they treat and which they refer to a specialist. The number of roots may be the determining factor in a decision concerning referral, or the key factor may be the chronic or acute status of the case. Others consider the complexity of the ultimate prosthesis as a factor in considering an endodontic referral. The most important variables in determining whether to refer a patient to a specialist are the skills of the clinician and the complexity of the case. The AAE developed guidelines for assessing endodontic case difficulty (available at https://www.aae.org/specialty/wp-content/uploads/sites/2/2017/10/2006casedifficultyassessmentformb_edited2010.pdf). The AAE Endodontic Case Difficulty Assessment Form enables a clinician to assign a level of difficulty to a particular case. The form describes cases with minimal, moderate, and high degrees of difficulty. This form lists criteria that can be used to identify cases that should be referred to a specialist. The use of surgical operating microscopes, endoscopes, and ultrasonics enables the specialist to predictably treat teeth that would not previously have been treatable.

ANXIETY

Anxiety presents a problem at many levels of dental care. Avoidance of dental treatment due to anxiety appears to be associated with significant deterioration of oral and dental health.[120] Even at the diagnostic stage, severe anxiety may confuse the process.[25] Several studies support the hypothesis that pain or fear of pain is a primary source of anxiety as well as an obstacle to seeking dental care.[50,120] Also, highly anxious patients appear to be more sensitive to pain.[26,49] High levels of anxiety have been found to negatively affect clinical procedures including local anesthesia.[64] In 2009, Binkley tested the hypothesis that having natural red hair color, which is caused by variants of the melanocortin-1 receptor *(MC1R)* gene, could predict a patient's experiencing dental care–related anxiety and dental care avoidance.[10] She found that participants with *MC1R* gene variants reported significantly more dental care–related anxiety and fear of dental pain than did participants with no *MC1R* gene variants. A more recent study confirmed the relationship between the red hair phenotype and anxiety. However, this study found no association between the red hair phenotype and the success of local anesthesia.[23]

It has been demonstrated that dental anxiety and expectation of pain had a profound effect on a patient's ability to understand information provided.[27] A person's cognitive ability to process information is significantly affected by stress.[27] A study found that 40% of patients who had minor oral surgery did not remember receiving both written and verbal instructions, contributing to 67% noncompliance with antibiotic prescriptions.[11] Patients' anxiety can compromise their understanding of complex treatment plans. Decisions made by a patient concerning options involving tooth retention or loss may be markedly affected by anxiety. In the field of medicine, there strong evidence to show that

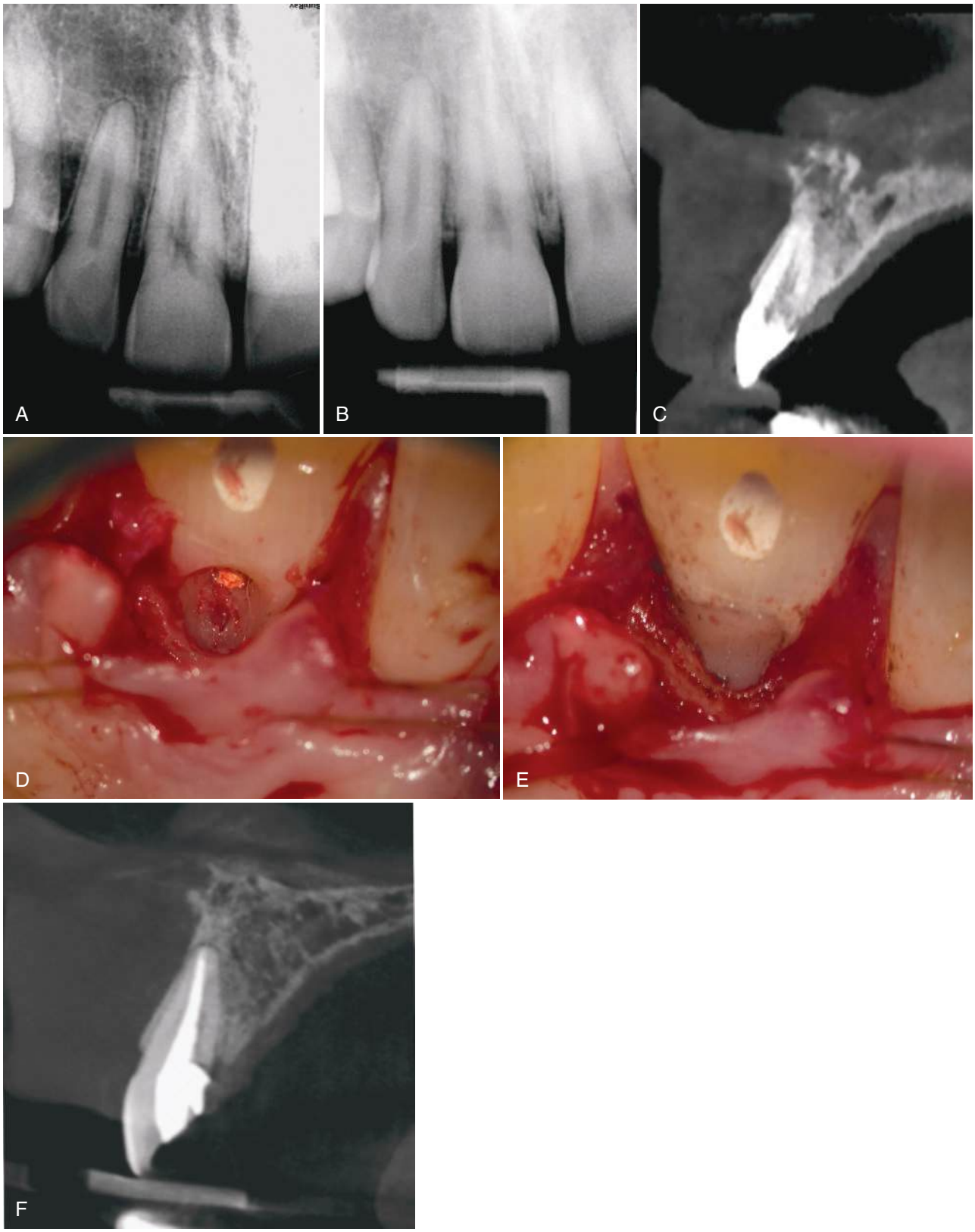


Fig. 5.14 A, Invasive cervical resorption of tooth #8 was detected on a periapical radiograph. **B,** A radiograph taken 9 years prior to the current examination. **C,** CBCT provided an accurate view of the location and the extent of the lesion. **D,** During surgery, following root canal therapy, the resorptive defect was excavated. **E,** The surgical site after restoration with Geristore. **F,** A 2-year follow-up CBCT shows that the lesion has not continued to extend.

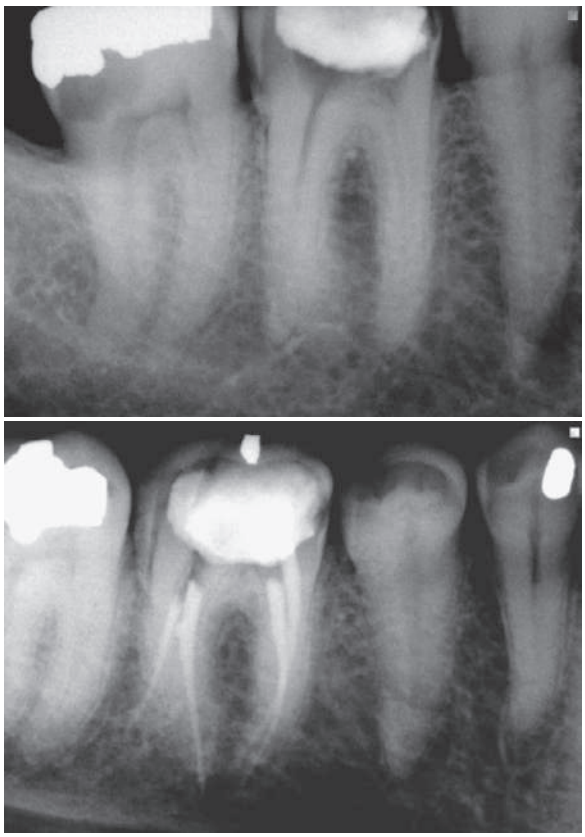


Fig. 5.15 The presence of curved roots and multiple canals is a complicating factor.

behavioral intervention for the highly anxious patient before treatment decreases anxiety before and after surgery, reduces posttreatment pain, and accelerates recovery.[45]

Unfortunately, the impact that a high level of anxiety can have on patient's cognition, local anesthesia, and intraoperative and postoperative experiences is not always recognized. A landmark medical study found that pretreatment discussion of surgical treatments and associated discomfort reduced by 50% the need for posttreatment morphine and reduced the time to discharge.[24] Existing research has focused primarily on the effect of pretreatment information on reducing anxiety and stress during surgery.[27]

Calm setting, reassurance by the clinician and explanation of the treatment plan, as well as a discussion about pain prevention strategies are all important steps even before treatment starts.[65] A written description as well as a verbal description of the proposed treatment are helpful. It may also be of value to have a family member or friend accompany the patient for a discussion of the treatment plan.

SCHEDULING CONSIDERATIONS

If a vital case is to be treated by a multivisit approach, it is suggested that the clinician allow 5 to 7 days between canal instrumentation and obturation to allow periradicular tissues to recover. When a vital case is to be treated in a single visit, adequate time must be scheduled so that the clinician can comfortably complete the procedure. Because profound inferior alveolar nerve block anesthesia may require approximately 15 to 20 minutes, it is wise to include that time when scheduling a patient's appointment (see also Chapter 6 on pain control and local anesthesia).

Appointments to fill nonvital cases should be scheduled approximately 1 week after instrumentation to maximize the antimicrobial effect of the intracanal dressing when calcium hydroxide is used.[2,99,100] Acute (pain or swelling) nonvital cases should be seen every 24 to 48 hours to monitor the patient's progress and bring the acute symptoms under control. Further cleaning and shaping are important components of the treatment as the clinician seeks to eliminate persistent microbes in the canal system. Long delays between visits contribute to the development of resistant microbial strains and should be avoided.

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